

DISTRIBUTION OF A QUADRATIC FORM IN NONCENTRAL NORMAL VECTORS USING GENERALISED LAGUERRE POLYNOMIALS

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SUMMARY

This paper gives a Laguerre type distribution of a positive definite quadratic form S in noncentral normal vectors by generalising the concept of Laguerre Polynomials. In some special cases, the distribution of the characteristic roots of S is established. Some important properties of the generalised Laguerre polynomials are established along with the explicit expressions of these polynomials up to degree two.

1. Introduction

The quadratic form in noncentral normal vectors is defined by $S = XAX'$ where A is an $n \times n$ positive definite (p.d.) matrix and the density function of a $p \times n$ matrix X is

$$\pi^{-pn/2} |\Sigma|^{-n/2} \text{etr}(-\Sigma^{-1}(X-\mu)(X-\mu)'/2)$$

with $\text{etr}(P) = \exp(\text{tr } P)$ and Σ being p.d. (i.e. column vectors of X are independently normal and this will be denoted by $N_{p,n}(\mu, \Sigma, I_n)$). When $\mu = 0$, the distribution of S was first given by Khatri (1966) in a (i) power-series type and (ii) mixture of Wishart distribution, and Shah (1970) extended this representation to the third type known as Laguerre series representation. For the central case, Khatri (1971) considered the convergence and the calculation of error terms for all these types of representation. When $\mu \neq 0$, the distribution of S is given by Crowther (1975) and Khatri (1975). The approaches given by them differ in presentation. Khatri (1975) extended the method of Kotz et. al. (1967) and Khatri (1971), giving some idea of error term, while Crowther (1975) extended Hayakawa's polynomials (1969, 1972) $P_K(T, A)$ to $P_K(T, A, B)$ and used them to get the distribution of S . Note that $P_K(T, A, I) = P_K(T, A)$. Here, we give the distribution of S to the third type of presentation, namely, Laguerre series representation. For this purpose, we generalised Laguerre polynomials $L_K^T(B)$ to $L_K^T(B, A)$ and $L_K^T(B, A, T)$ and establish the distribution of S in terms of these polynomials. Extensions of these

polynomials are done using $P_{\kappa}(T, A, B)$ polynomials in the sense of Constantine (1966, p. 218). Some properties of these polynomials are given.

2. Polynomials $L_{\kappa}^T(B, A, T)$ and $P_{\kappa}(T, A, B)$

Crowther (1975) defined the polynomials $P_{\kappa}(T, A, B)$ through the relation:

$$(2.1) P_{\kappa}(T, A, B) = \pi^{-pn/2} \text{etr}(TT') \int_U \text{etr}(-UU' - 2iTU') C_{\kappa}(-B U A U') dU \text{ where}$$

$C_{\kappa}(P)$ is a zonal polynomial of degree k in P (for definition see James (1964)),

$$\kappa = \{k_1, \dots, k_p\}, k_1 \geq k_2 \geq \dots \geq k_p \geq 0 \text{ and } \sum_{j=1}^p k_j = k, i = \sqrt{-1}, B \text{ and } A \text{ are sym-}$$

metric matrices of order $p \times p$ and $n \times n$ respectively and T is a $p \times n$ complex matrix. This can be defined by

$$(2.2) P_{\kappa}(T, A, B) = E_V \{C_{\kappa}(-B(V - iT)A(V - iT)')\}$$

where $V \sim N_{p,n}(0, I, 1/2)$. Let us define the generalised Bessel function as (see for example Herz (1955) or Constantine (1966)),

$$(2.3) A_{\tau}(R) = \sum_{k=0}^{\infty} \sum_{\kappa} C_{\kappa}(-R) / \Gamma_p(\tau + \frac{1}{2}(p+1), \kappa) k!$$

where R is a $p \times p$ symmetric matrix and

$$(2.4) \Gamma_p(b, \kappa) = \pi^{p(p-1)/4} \prod_{j=1}^p \Gamma(b + k_j - \frac{j-1}{2}) = \Gamma_p(b) (b)_{\kappa}$$

$$\Gamma_p(b) = \Gamma_p(b, 0).$$

If $A = I_n$ in (2.1) and $\alpha = (n-p-1)/2$, then integrating over U such that $UU' = R$, (2.1) can be written as

$$(2.5) P_{\kappa}(T, I, B) = \text{etr}(TT') \int_{R \geq 0} \text{etr}(-R) |R|^{\alpha} C_{\kappa}(-BR) A_{\alpha}(TT'R) dR.$$

Hence, noting the definition of Constantine (1966, p. 218) for Laguerre polynomials, we get

$$(2.6) P_{\kappa}(T, I_n, I_p) = (-1)^k L_{\kappa}^{\alpha}(TT').$$

Thus, (2.5) becomes a generalization of Laguerre polynomials. Hence, let us define a generalised Laguerre polynomial $L_k^T(S, B)$ for any $p \times p$ symmetric matrices S and B as

$$(2.7) \quad L_k^T(S, B) \operatorname{etr}(-S) = \int_{R \geq 0} \operatorname{etr}(-R) |R|^T C_k(BR) A_T(SR) dR.$$

Hence, noting (2.5) and (2.7), we have

$$P_k(T, I, B) = (-1)^k L_k^\alpha(TT', B) \text{ with } \alpha = (n-p-1)/2.$$

Further, let us write in (2.7), $B = (V-iT)A(V-iT)'$, and then integrate over V which is distributed as $N_{p,n}(0, I, I/2)$. Then in the spirit of the definition of Laguerre polynomials, we define a generalised Laguerre polynomial $L_k^T(S, A, T)$ through

$$(2.8) \quad L_k^T(S, A, T) = \operatorname{etr}(S) \int_{R \geq 0} \operatorname{etr}(-R) |R|^T P_k(T, -A, +R) A_T(SR) dR$$

or

$$(2.9) \quad L_k^T(S, A, T) = E_V(L_k^T(S, (V-iT)A(V-iT)'))$$

where S and A are $p \times p$ and $n \times n$ real symmetric matrices and T is a $p \times n$ complex matrix. If H is any $p \times p$ orthogonal matrix and S is replaced by HSH' in (2.8), then it is easy to see that

$$(2.10) \quad L_k^T(HSH', A, T) = L_k^T(S, A, HT),$$

and if H_1 is an $n \times n$ orthogonal matrix, then

$$(2.11) \quad L_k^T(S, H_1 A H_1', T) = L_k^T(S, A, T H_1).$$

Noting the result

$$(2.12) \quad \int_{O(p)} P_k(T, A, H R H') dH = C_k(R) P_k(T, A) / C_k(I_p),$$

we get from (2.8),

$$(2.13) \quad \int_{O(p)} L_k^T(HSH', A, T) dH = P_k(T, -A) L_k^T(S) / C_k(I_p).$$

Using the definition (2.7), the Laplace transform of $|S|^T L_k^T(S, B)$ is given by

$$(2.14) \quad g_k^T(Z, B) = \int_{S \geq 0} \operatorname{etr}(-ZS) |S|^T L_k^T(S, B) dS \text{ for } \operatorname{Re}(Z) > I,$$

$$= \int_{S>0} \text{etr}(-(Z-I)S) |S|^{\tau} \int_{R>0} \text{etr}(-R) |R|^{\tau} C_{\kappa}(BR) A_{\tau}(SR) dR dS.$$

Interchanging the integrals and using

$$\int_{S>0} \text{etr}(-(Z-I)S) |S|^{\tau} A_{\tau}(SR) dS = |Z-I|^{-\tau-(p+1)/2} \text{etr}(-(Z-I)^{-1}R),$$

(2.14) can be written as

$$\begin{aligned} (2.15) \quad g_{\kappa}^{\tau}(Z, B) &= |Z-I|^{-\tau-(p+1)/2} \int_{R>0} \text{etr}(-(I+(Z-I)^{-1})R) |R|^{\tau} C_{\kappa}(BR) dR \\ &= \Gamma_p(\tau+(p+1)/2, \kappa) |Z|^{-\tau-(p+1)/2} C_{\kappa}((I-Z^{-1})B) \end{aligned}$$

for $\text{Re}(Z) > I$. Hence, using (2.9) and (2.15), the Laplace transform of $|S|^{\tau} L_{\kappa}^{\tau}(S, A, T)$ is given by

$$\begin{aligned} (2.16) \quad \int_{S>0} \text{etr}(-ZS) |S|^{\tau} L_{\kappa}^{\tau}(S, A, T) dS \\ = \Gamma_p(\tau+(p+1)/2, \kappa) |Z|^{-\tau-(p+1)/2} P_{\kappa}(T, -A, I-Z^{-1}) \end{aligned}$$

for $\text{Re}(Z) > I$. Using the inverse Laplace transform, we get

$$\begin{aligned} (2.17)' \quad L_{\kappa}^{\tau}(S, A, T) &= \Gamma_p(\tau+(p+1)/2, \kappa) [2^{p(p-1)/2}/(2\pi i)^{p(p+1)/2}] \\ &\quad \int_{\text{Re}(Z)>I} \text{etr}(ZS) |Z|^{-\tau-(p+1)/2} |S|^{\tau} P_{\kappa}(T, -A, I-Z^{-1}) dZ \end{aligned}$$

or (2.17)' can be written as

$$\begin{aligned} (2.17) \quad L_{\kappa}^{\tau}(S, A, T) &= \Gamma_p(\tau+(p+1)/2, \kappa) [2^{p(p-1)/2}/(2\pi i)^{p(p+1)/2}] \\ &\quad \int_{\text{Re}(Z)>I} \text{etr}(Z) |Z|^{-\tau-(p+1)/2} P_{\kappa}(T, -A, I-S^{1/2}Z^{-1}S^{1/2}) dZ \end{aligned}$$

where $S^{1/2}$ is a symmetric square root of S , ($S^{1/2}$ is allowed to be a complex matrix) and (2.17) is valid even when S is singular. From (2.17), we get

$$(2.18) \quad L_{\kappa}^{\tau}(O, A, T) = (\tau+(p+1)/2)_{\kappa} P_{\kappa}(T, -A)$$

or (2.18) can be obtained from (2.8) because

$$A_{\tau}(O) = \{ \Gamma_p(\tau + (p+1)/2) \}^{-1}.$$

Further, noting that

$$P_k(O, -A, B) = (n/2)_k C_k(A) C_k(B) / C_k(I_n)$$

in (2.17) or (2.8), we get

$$(2.19) \quad L_k^T(S, A, O) = (n/2)_k C_k(A) L_k^T(S) / C_k(I_n).$$

Using the definition $L_k^T(S, B)$, we have

$$\Sigma_k L_k^T(S, B) = \text{etr}(S) \int_{R>0} \text{etr}(-R) |R|^{\tau} (\text{tr } BR)^k A_{\tau}(SR) dR$$

and hence the generating function of $\Sigma_k L_k^T(S, B)$ is given by

$$(2.20) \quad \sum_{k=0}^{\infty} \theta^k \Sigma_k L_k^T(S, B) / k! \\ = |I - \theta B|^{-\tau - (p+1)/2} \text{etr}(-\theta SB(I - \theta B)^{-1}).$$

When $p=1$ in (2.20), we get

$$(2.21) \quad L_k^T(s, b) = b^k L_k^T(s)$$

and hence

$$(2.22) \quad L_k^T(s, A, t) = \alpha_k L_k^T(s)$$

where the generating function of α_k is

$$(2.23) \quad \sum_{k=0}^{\infty} \alpha_k \theta^k / k! = |I - \theta A|^{-1/2} \exp(-\theta t A (I - \theta A)^{-1} t').$$

From (2.23), we can obtain the α_k 's easily and thus calculate $L_k^T(s, A, t)$ for $p=1$ and for any k . For any p , expressions of $L_k^T(S, B)$ and $L_k^T(S, A, T)$ for $k=1$ and 2 only, are given in the following section.

To establish the orthogonality property of the generalised Laguerre polynomials $L_k^T(S, A)$, we first establish the generating function of $L_k^T(S, A)$:

Let us define

$$\begin{aligned}
 (2.24) \quad h(S, A, Z) &= \sum_{k=0}^{\infty} \sum_{\kappa} L_k^T(S, A) C_{\kappa}(Z)/k! C_{\kappa}(I) \\
 &= \text{etr}(S) \int_{R \geq 0} \text{etr}(-R) |R|^T A_{\tau}(SR) g(R, A, Z) dR
 \end{aligned}$$

where

$$\begin{aligned}
 (2.25) \quad g(R, A, Z) &= \sum_{k=0}^{\infty} \sum_{\kappa} C_{\kappa}(AR) C_{\kappa}(Z)/k! C_{\kappa}(I_p) \\
 &= \sum_{k=0}^{\infty} \sum_{\kappa} (k!)^{-1} \int_{O(p)} C_{\kappa}(RA^{1/2} HZH'A^{1/2}) dH \\
 &= \int_{O(p)} \text{etr}(RA^{1/2} HZH'A^{1/2}) dH,
 \end{aligned}$$

$A^{1/2}$ being at least a complex symmetric matrix. Using (2.25) in (2.24) and interchanging the integration, we get

$$(2.26) \quad h(S, A, Z) = \int_{O(p)} |I - A^{1/2} HZH'A^{1/2}|^{-\tau - 1/2(p+1)} \text{etr}(S - S(I - A^{1/2} HZH'A^{1/2})^{-1}) dH.$$

Now, for the orthogonality property, we note the following result:

$$\begin{aligned}
 (2.27) \quad &\int_{S \geq 0} \text{etr}(-S) |S|^T h(S, A, Z) L_{\kappa}^T(S, B) dS \\
 &= \int_{O(p)} dH \int_{S \geq 0} |I - A^{1/2} HZH'A^{1/2}|^{-\tau - 1/2(p+1)} \text{etr}(-S(I - A^{1/2} HZH'A^{1/2})^{-1}) |S|^T \\
 &\quad \cdot L_{\kappa}^T(S, B) dS \\
 &= \int_{O(p)} \Gamma_p(\tau + (p+1)/2, \kappa) C_{\kappa}(HZH'A^{1/2} B A^{1/2}) dH \text{ using (2.15)} \\
 &= \Gamma_p(\tau + (p+1)/2, \kappa) C_{\kappa}(AB) C_{\kappa}(Z)/C_{\kappa}(I_p).
 \end{aligned}$$

Hence, using the definition of $h(S, A, Z)$ given in (2.24), we get the following important orthogonality property:

$$(2.28) \quad \int_{S \geq 0} \text{etr}(-S) |S|^T L_{\kappa}^T(S, A) L_{\kappa}^T(S, B) dS = k! \Gamma_p(\tau + (p+1)/2, \kappa) C_{\kappa}(AB)$$

and

$$(2.29) \int_{S>0} \text{etr}(-S) |S|^T L_k^T(S, A) L_\eta^T(S, B) dS = 0 \text{ if } k \neq \eta.$$

From (2.28) and (2.29), similar results for $L_k^T(S, A, T)$ can be established.

3. Expressions for $L_k^T(S, B)$ and $L_k^T(S, A, T)$ up to $k=2$

Using (2.7) and (2.9) we shall establish these relations. Crowther (1975) has given expressions of $P_k(T, A, B)$ up to $k=3$ and these are given up to $k=2$ as follows:

$$P_{(0)}(T, A, B) = 1, P_{(1)}(T, I, B) = \text{tr}(BTT') - (n/2)\text{tr}B,$$

$$P_{(2)}(T, I, B) = (2/3) [E(\text{tr}BUU')^2 - E \text{tr}(BUU')^2]$$

and

$$P_{(12)}(T, I, B) = [E(\text{tr}BUU')^2 - 2E \text{tr}(BUU')^2]/3$$

where

$$E(\text{tr}BUU')^2 = (n/2) \text{tr} B^2 + (n/2)^2 (\text{tr} B)^2 - n \text{tr} B \text{tr} BTT' - 2 \text{tr} B^2 TT' + \text{tr} BTT')^2$$

and

$$E \text{tr}(BUU')^2 = [n(n+1)/4] \text{tr} B^2 + (n/4) (\text{tr} B)^2 - (n+1) \text{tr} B^2 TT' - \text{tr} B \text{tr} BTT' + \text{tr} (BTT')^2.$$

Hence, the expressions for $L_k^T(S, B)$ can be obtained from the above after replacing B by $-B$, n by $(2\tau+p+1)$ and TT' by S . Thus, we have

$$L_{(0)}^T(S, B) = 1, L_{(1)}^T(S, B) = (\text{tr} B)(\tau+(p+1)/2) - \text{tr} BS,$$

$$L_{(2)}^T(S, B) = 2(\delta_1 - \delta_2)/3 \text{ and } L_{(12)}^T(S, B) = (\delta_1 - 2\delta_2)/3$$

where

$$\delta_1 = (\tau+(p+1)/2) \text{tr} B^2 + (\tau+(p+1)/2)^2 (\text{tr} B)^2 - (2\tau+p+1) \text{tr} B \text{tr} SB - 2 \text{tr} B^2 S + (\text{tr} BS)^2$$

$$\text{and } \delta_2 = (2\tau + p + 1)(2\tau + p + 2)(\text{tr } B^2/4) - (2\tau + p + 2) \text{tr } B^2 S \\ + \frac{1}{2} (\tau + (p + 1)/2)(\text{tr } B)^2 - \text{tr } B \text{tr } SB + \text{tr } (SB)^2.$$

Hence, using (2.9), we get

$$L_{(0)}^{\tau}(S, A, T) = 1$$

$$L_{(1)}^{\tau}(S, A, T) = (\tau + (p + 1)/2)((p/2) \text{tr } A - \text{tr } ATT') - (\text{tr } A \text{tr } S/2) + \text{tr } STAT'$$

$$L_{(2)}^{\tau}(S, A, T) = 2(\delta_1^{(1)} - \delta_2^{(2)})/3 \text{ and}$$

$$L_{(1^2)}^{\tau}(S, A, T) = (\delta_1^{(1)} - 2\delta_2^{(2)})/3$$

where $\delta_1^{(1)}$ and $\delta_2^{(2)}$ are obtained from δ_1 and δ_2 by replacing

$$(i) \quad \text{tr } SB \text{ by } (\text{tr } A)(\text{tr } S/2) - \text{tr } STAT',$$

$$\text{tr } B \text{ by } (p/2) \text{tr } A - \text{tr } ATT',$$

$$(ii) \quad \text{tr } B^2 S \text{ by } \frac{1}{4} \text{tr } S (\text{tr } A)^2 + \frac{1}{4} (p + 1)(\text{tr } S)(\text{tr } A^2)$$

$$- (\text{tr } A)(\text{tr } STAT') - ((p + 2)/2) \text{tr } STA^2 T'$$

$$- \frac{1}{2} (\text{tr } S)(\text{tr } TA^2 T') + \text{tr } (TAT')^2 S,$$

$$(iii) \quad \text{tr } (BS)^2 \text{ by } \frac{1}{4} (\text{tr } A^2)(\text{tr } S^2) + \frac{1}{4} (\text{tr } A)^2 (\text{tr } S^2)$$

$$+ \frac{1}{4} (\text{tr } A^2)(\text{tr } S)^2 - \text{tr } S^2 TA^2 T'$$

$$- (\text{tr } A)(\text{tr } S^2 TAT') - (\text{tr } S)(\text{tr } STA^2 T')$$

$$+ \text{tr } (STAT')^2$$

4. Distribution of a quadratic form $S = XAX'$.

Let $A = H_2 D_{\alpha} H_2'$, H_2 being an orthogonal matrix and $D_{\alpha} = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_n)$. Let $\mu H_2 = \nu = (\nu_1, \dots, \nu_n)$. Then, the Laplace transform of the density function of S is given by

$$(4.1) \quad E \text{etr}(-ZS) = \prod_{j=1}^n \{ |I + 2\alpha_j \Sigma Z|^{-1/2} \exp(-\alpha_j \nu_j' Z (I + 2\alpha_j \Sigma Z)^{-1} \nu_j) \}$$

(see Khatri (1971) or (1975)). Putting $\Sigma^{1/2} Z \Sigma^{1/2} = q(\theta - I)$, $q \neq 0$,

$$b_{j(1)} = 1 - q\alpha_j, \quad b_{j(2)} = (q\alpha_j)^{-1} - 1,$$

$$w_{j(1)} = (2\Sigma)^{-1/2} v_j / \sqrt{b_{j(2)}} \text{ and } w_{j(2)} = (2\Sigma)^{-1/2} v_j / \sqrt{b_{j(1)}}$$

for $j=1, 2, \dots, n$ in (4.1), one can rewrite (4.1) as

$$(4.2) \quad E \operatorname{etr}(-ZS) = |\theta|^{n/2} M_k(\theta) \text{ for } k=1, 2$$

where

$$(4.3) \quad M_1(\theta) = \prod_{j=1}^n \{ |I + b_{j(1)}(\theta - I)|^{-1/2} \exp(b_{j(1)} w'_{j(1)} (\theta - I)(I + b_{j(1)}(\theta - I))^{-1} w_{j(1)}) \} \\ = E_V \{ \operatorname{etr}(-(\theta - I)(V - iW_1)B_1(V - iW_1)') \}$$

and

$$(4.4) \quad M_2(\theta) = |qA|^{-p/2} \operatorname{etr}(-\Sigma^{-1} \mu \mu' / 2) \prod_{j=1}^n |I + b_{j(2)} \theta|^{-1/2} \\ \exp\left(\sum_{j=1}^n b_{j(2)} w'_{j(2)} \theta (I + b_{j(2)} \theta)^{-1} w_{j(2)}\right) \\ = |qA|^{-p/2} \operatorname{etr}(-\Sigma^{-1} \mu \mu' / 2) E_V \{ \operatorname{etr}(-\theta(V - iW_2)B_2(V - iW_2)') \}$$

with $V \sim N_{p,n}(0, I_p/2, I_n)$, $B_1 = I - qA$, $B_2 = (qA)^{-1} - I$,

$$W_1 = (2\Sigma)^{-1/2} \mu B_2^{-1/2} \text{ and } W_2 = (2\Sigma)^{-1/2} \mu B_1^{-1/2}.$$

Notice that W_1 and W_2 may be complex matrices on account of the square roots of symmetric matrices B_1 and B_2 .

Using (2.2) and $\operatorname{etr}(T) = \sum_{k=0}^{\infty} \Sigma_k C_k(T)/k!$ in (4.2), we get

$$(4.5) \quad E \operatorname{etr}(-ZS) = |\theta|^{n/2} \sum_{k=0}^{\infty} \Sigma_k P_k(W_1, B_1, \theta - I)/k!$$

and

$$(4.6) \quad E \operatorname{etr}(-ZS) = |\theta|^{n/2} |qA|^{-p/2} \operatorname{etr}(-\Sigma^{-1} \mu \mu' / 2) \sum_{k=0}^{\infty} \Sigma_k P_k(W_2, B_2, \theta)/k!$$

Let us denote

$$(4.7) \quad W_p(n, \Sigma_1; S) = |2\Sigma_1|^{-n/2} \{\Gamma_p(n/2)\}^{-1} |S|^{(n-p-1)/2} \text{etr}(-\Sigma_1^{-1} S/2).$$

Then, from (2.16), the Laplace transform of

$$W_p(n, \Sigma/q; S) \{(n/2)_k\}^{-1} L^{(n-p-1)/2}(q\Sigma^{-1/2} S \Sigma^{-1/2}/2, B_1, W_1)$$

is

$$|\theta|^{n/2} P_k(W_1, B_1, \theta - I)$$

and using (2.1), the Laplace transform of

$$W_p(n, \Sigma/q; S) \{(n/2)_k\}^{-1} P_k(W_2, B_2, q\Sigma^{-1/2} S \Sigma^{-1/2}/2) \text{ is}$$

$$|\theta|^{n/2} P_k(W_2, B_2, \theta).$$

Using the above results and the uniqueness property of the Laplace transform in (4.5) and (4.6), we get the density function of S as

$$(4.8) \quad W(n, \Sigma/q; S) \sum_{k=0}^{\infty} \Sigma_k \{(n/2)_k k!\}^{-1} L^{(n-p-1)/2}(q\Sigma^{-1/2} S \Sigma^{-1/2}/2, B_1, W_1)$$

and

$$(4.9) \quad |qA|^{-p/2} \text{etr}(-\Sigma^{-1} \mu \mu'/2) W_p(n, \Sigma/q; S)$$

$$\sum_{k=0}^{\infty} \Sigma_k \{(n/2)_k k!\}^{-1} P_k(W_2, B_2, q\Sigma^{-1/2} S \Sigma^{-1/2}/2).$$

The expression (4.9) was established by Crowther (1975), while the expression (4.8) generalises the results of Shah (1970) and Khatri (1971) to noncentral variates. The convergence of (4.8) and (4.9) depends on a choice of q and the proper value of q can be found using a similar technique given by Khatri (1975). When $A=I$ in (4.8) and $q>1$, we have

$$\begin{aligned} & L^{(n-p-1)/2}(q\Sigma^{-1/2} S \Sigma^{-1/2}/2, (1-q)I, (2\Sigma)^{-1/2} \mu(q/(1-q)^{1/2})) \\ & \rightarrow L^{(n-p-1)/2}(\Sigma^{-1/2} S \Sigma^{-1/2}/2, -\Sigma^{-1/2} \mu \mu' \Sigma^{-1/2}/2) \quad \text{as } q \rightarrow 1. \end{aligned}$$

and hence, the density function of S (i.e. the noncentral Wishart density function) can be written as

$$(4.10) W_p(n, \Sigma; S) \sum_{k=0}^{\infty} \sum_{\kappa} \{(n/2)_{\kappa} k!\}^{-1} L_k^{(n-p-1)/2} (\Sigma^{-1/2} S \Sigma^{-1/2} / 2, -\Sigma^{-1/2} \mu \mu' \Sigma^{-1/2} / 2).$$

When $p=1$, (4.8) gives the density function of s as

$$(4.11) W_1(n, \sigma^2/q; s) \sum_{k=0}^{\infty} \{(n/2)_k k!\}^{-1} d_k L_k^{(n-2)/2} (qs/2\sigma^2)$$

where the generating function of d_k is given by

$$(4.12) \sum_{k=0}^{\infty} d_k \theta^k / k! = |I - \theta(I - qA)|^{-1/2} \exp(-\theta q \mu A (I - \theta(I - qA))^{-1} \mu' / 2\sigma^2)$$

with μ being a row vector with n elements. (4.11) is given by Kotz et. al (1967) and the choice of q suggested by them can be used in (4.8) without any difficulty. From (4.11), the distribution of s can be given by

$$(4.13) F(x) = P(s \leq x) \\ = I(n; qx/\sigma^2) + \sum_{k=0}^{\infty} \frac{d_{k+1} W_1(n+2, \sigma^2/q; x)}{(k+1)! (\frac{1}{2}n+1)_k (q/2\sigma^2)} L_k^{n/2} (qx/2\sigma^2)$$

where $I(n; 2y) = \int_0^y z^{(n-2)/2} \exp(-z) dz / \Gamma(n/2)$, the central chi-square distribution function with n degrees of freedom.

5. The distribution of the characteristic roots of S and the symmetrized density of S

The density function of the characteristic roots, $\lambda_1 > \lambda_2 > \dots > \lambda_p > 0$, of S is given by Crowther (1975) in the power-series form which is a slowly convergent series and this is true for the distribution of λ_1 established by him. To obtain the density function of $\lambda_1 > \lambda_2 > \dots > \lambda_p > 0$ in a rapidly convergent series, we assume $\Sigma = I$ (or $\sigma^2 I = \Sigma$) in (4.8) for which some results are given by Hayakawa (1972) for $q=1$. The constant q is chosen in such a way as to improve convergence of the series. We use the transformation $S = H\Delta H'$ where H is an orthogonal matrix and $\Delta = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$. The Jacobian of the transformation is

$$(5.1) J(S; H, \Delta) = \alpha_p(\Delta) \pi^{1/2} p^2 \{\Gamma_p(p/2)\}^{-1} dH$$

where dH is a unit invariant Haar measure over the orthogonal space $O(p)$ and

$$\alpha_p(\Delta) = \prod_{i=1}^{p-1} \prod_{j=i+1}^p (\lambda_i - \lambda_j).$$

Using this in (4.8) and integrating over H with the help of (2.13), we get the density function of $\lambda_1 > \dots > \lambda_p > 0$ when $\Sigma = I$ as

$$(5.2) \quad \pi^{1/2} p^2 \{\Gamma_p(p/2)\}^{-1} W_p(n, I/q; \Delta) \alpha_p(\Delta)$$

$$\sum_{k=0}^{\infty} \sum_{\kappa} \{(n/2)_{\kappa} C_{\kappa}(I_p) k!\}^{-1} P_{\kappa}(W_1, qA-I) L_{\kappa}^{(n-p-1)/2}(q\Delta/2)$$

where $W_1 = \mu(I - qA)^{-1/2} (qA)^{1/2}/\sqrt{2}$ and $q > 0$ such that $(I - qA)$ is nonsingular.

In the special case when $A = I$, with the help of (2.6) and (2.12), we can write the density function of $\lambda_1 > \dots > \lambda_p > 0$ as

$$(5.3) \quad \pi^{1/2} p^2 \{\Gamma_p(p/2)\}^{-1} W_p(n, I/q; \Delta) \alpha_p(\Delta)$$

$$\sum_{k=0}^{\infty} \sum_{\kappa} \{(n/2)_{\kappa} C_{\kappa}(I_p) k!\}^{-1} (1-q)^k L_{\kappa}^{(n-p-1)/2}(q\Omega/2(1-q)) L_{\kappa}^{(n-p-1)/2}(q\Delta/2)$$

where $\Omega = \mu\mu'/2$, $q > 0$ and $q \neq 1$. Notice that

$$(5.4) \quad \lim_{q \rightarrow 1} (1-q)^k L_{\kappa}^{(n-p-1)/2}(q\Omega/2(1-q)) = C_{\kappa}(-\Omega),$$

and at $q=1$, we get the same expression as that given by Roux and Raath (1973). Hence, in the sense of Roux and Raath (1975), the symmetrized density function of S is given by

$$(5.5) \quad W_p(n, I/q; S) \sum_{k=0}^{\infty} \sum_{\kappa} \{(n/2)_{\kappa} C_{\kappa}(I_p) k!\}^{-1} P_{\kappa}(W_1, qA-I) L_{\kappa}^{(n-p-1)/2}(qS/2),$$

which gives the result of Roux and Raath (1975) as a special case when $A = I$ and $q \rightarrow 1$.

Remark:

Similar results for the distribution of Hermitian forms in complex normal vectors will be published in a later communication.

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